Flipkart Performance and Sentiment Analysis (FPSA)

End-to-end MySQL analytics on 66k+ products analyzing revenue trends, pricing elasticity, and quality risks.

Author: Shiv Dixit
Roll Number: 2401220100217
GitHub Repository: https://github.com/shiv9956/Flipkart-Performance-Sentiment-Analytics

Step 1 - Raw Data Ingestion & Schema Exploration

Sourcing the raw e-commerce dataset (`uncleaned.csv`) containing over 66,000 unvalidated catalog listings from Kaggle.

Identifiers	Financials	Metrics	Logistics & Metadata
`product_id, product_name, category, brand`	`price, discount_percent, final_price`	`rating, review_count, stock_available, units_sold, product_score, seller_rating`	`seller, seller_city, delivery_days, weight_g, warranty_months, color, size, return_policy_days, is_returnable, payment_modes, shipping_weight_g, listing_date`

product_id	product_name	category	brand
FKP0000001	Adidas Ultra 664	Toys	Adidas
FKP0000002	LG Series 124	Fashion	LG
FKP0000003	Redmi Model 35	Beauty	Redmi
FKP0000004	Sony Edition 769	Toys	Sony
FKP0000005	Boat Prime 291	Home & Kitchen	Boat

Step 2 - Development Environment & Dependencies

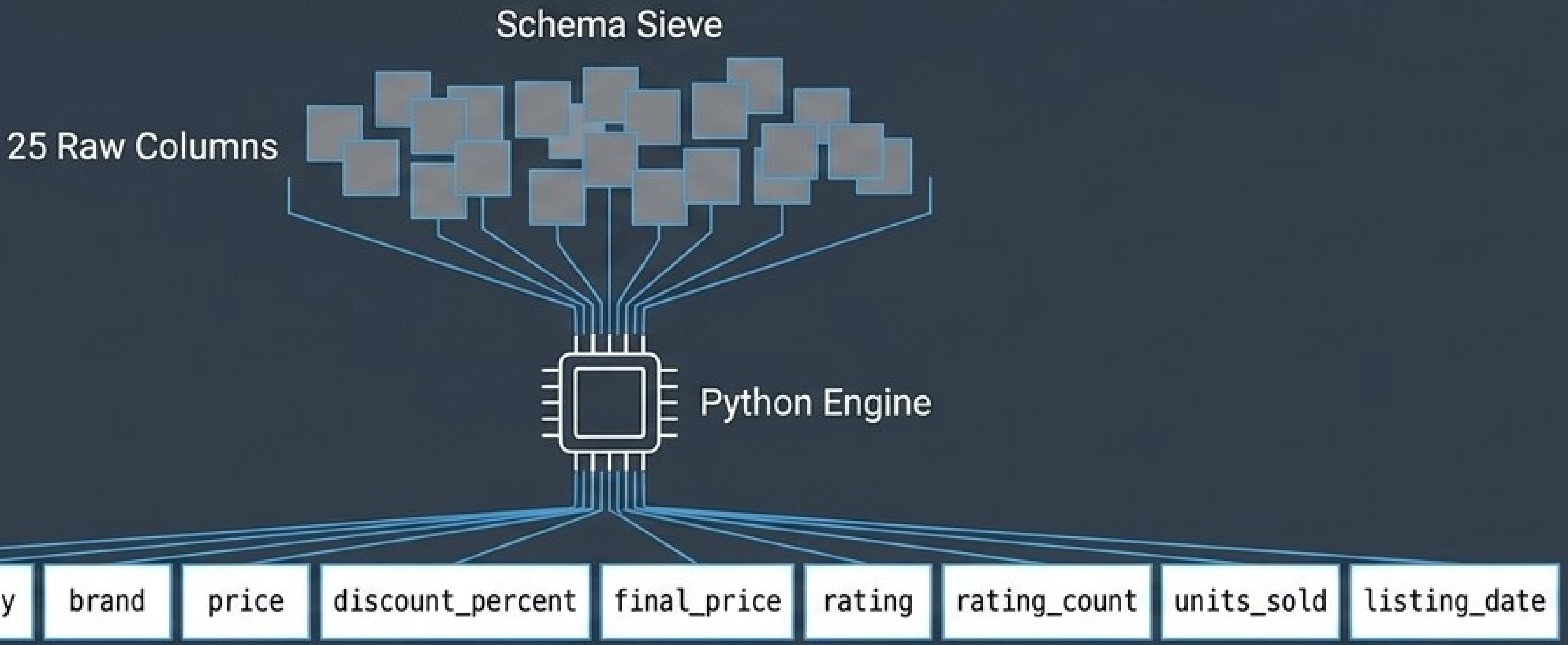
Establishing the Python data processing environment via Jupyter Notebook / Google Colab for robust memory management of the 66k+ row dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
from sqlalchemy import create_engine
```

price	discount_percent	final_price	rating	rating_count
35547.34	15	30215.24	1.9	26975
30693.79	10	27624.41	3.2	45848
51214.50	40	30728.70	4.4	5138
33168.49	30	23217.94	2.8	2000
9330.32	5	8863.80	3.6	9135

Step 3 - Data Cleaning & Feature Filtering

```python
# Drop metadata noise
df.drop(columns=['seller_city', 'shipping_weight_g', 'payment_modes', ...], inplace=True)
# Handle Nulls
df['rating'].fillna(df['rating'].median(), inplace=True)
df['rating_count'].fillna(0, inplace=True)
# Feature Engineering
df['final_price'] = df['price'] * (1 - df['discount_percent'] / 100)
```



<table>
<tr><th>Dirty</th><th>Clean</th></tr>
<tr><td>FKP0030096 | Toys | Dell | NaN | 20</td><td>FKP0030096 | Toys | Dell | 4.2 | 20</td></tr>
</table>

Step 4 - Exploratory Data Analysis & Descriptive Statistics

Execution of univariate statistical checks on the newly generated df_clean dataframe to validate distributions.

```
df_clean.info()
```

```
df_clean.describe().round(2)
```

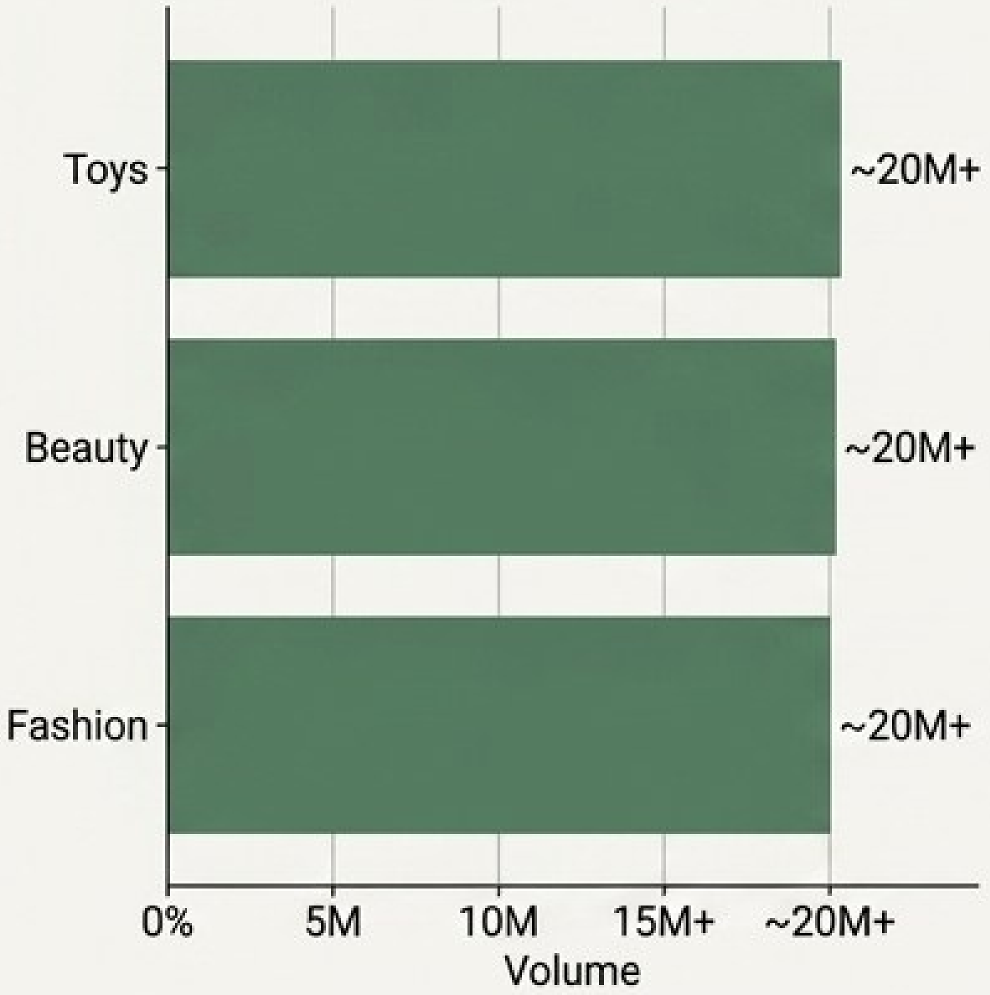
```
df_clean.groupby('category').agg({'units_sold':
'sum'})
```

price	Mean ₹30,123.00 (Standardized)
discount_percent	Mean ~21.4% (Consistent baseline)
units_sold	Max scaling to high volumes (20M+ per top category)
rating	Range safely from 1.0 to 5.0 (Nulls imputed)

Validation Check: Zero null values remaining across 66,516 rows.

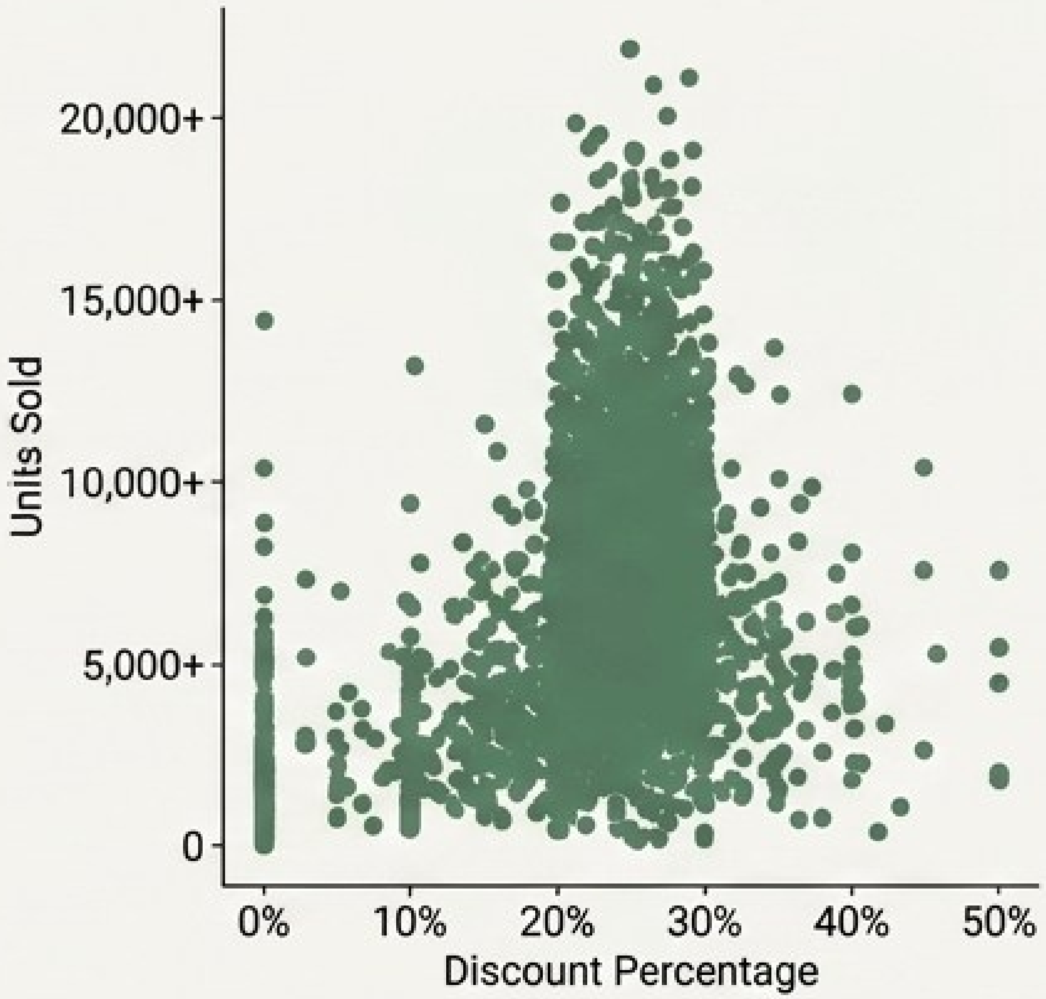
Step 5 - Static Analytical Visualizations

Volume



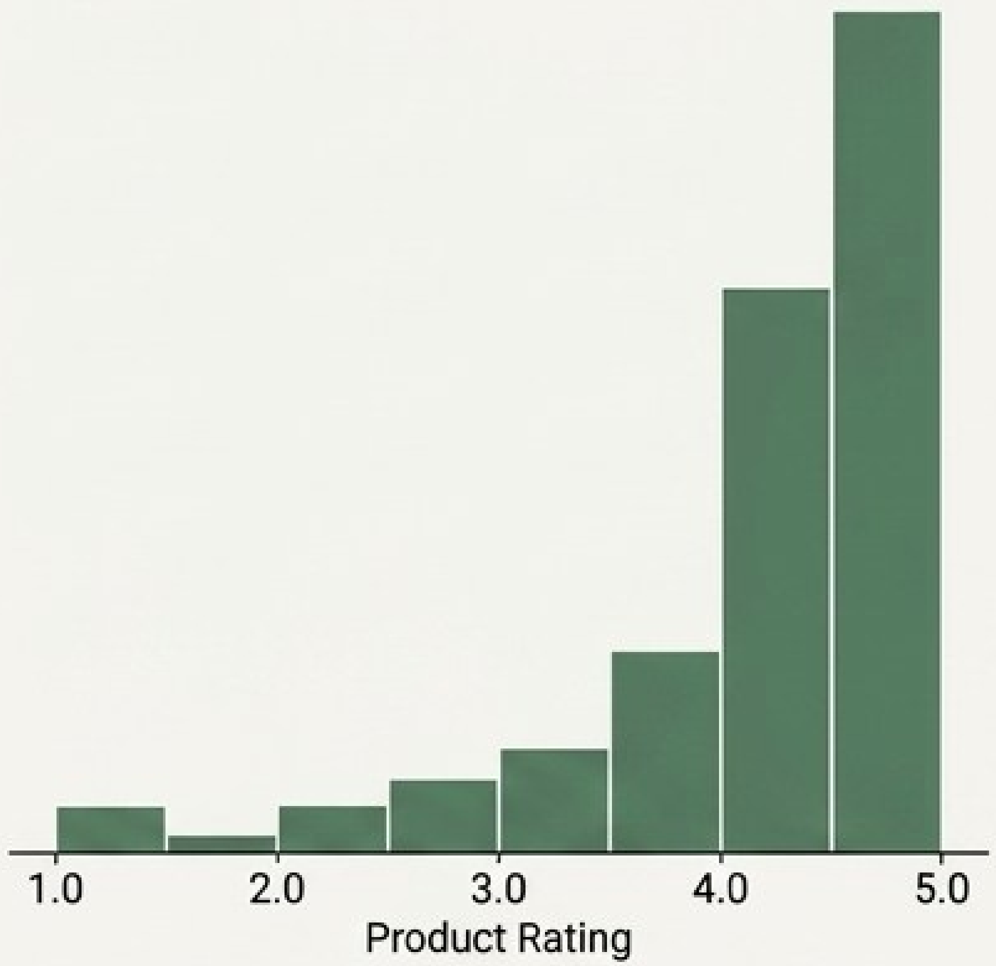
Toys, Beauty & Fashion lead marketplace volume.

Elasticity



Sales density peaks at 20-30% discount mark.

Sentiment



Ratings strongly skew positive post-imputation.

Step 6 - Relational Database Pipeline & Schema Setup

Migrating the static CSV into an enterprise-grade relational MySQL database (flipkart_db) for complex, multi-layered querying.

```
engine = create_engine('mysql+pymysql://user:pass
@localhost/flipkart_db')
```

```
CREATE TABLE flipkart_cleaned (
  product_id VARCHAR(50) PRIMARY KEY,
  category VARCHAR(100) NOT NULL,
  price DECIMAL(10, 2) NOT NULL,
  -- (Abbreviated for display)
);
CREATE INDEX idx_category ON
flipkart_cleaned(category);
```

```
Query OK, 0 rows affected (0.00 sec)

LOAD DATA LOCAL INFILE 'flipkart-cleaned.csv'
INTO TABLE flipkart_cleaned...
Query OK, 66516 rows affected (2.59 sec)
Records: 66516 Deleted: 0 Skipped: 0 Warnings: 0
```

Step 7 - Advanced SQL Business Queries (Part 1)

Query-to-Insight Card - Category Dominance

```
SELECT category, SUM(units_sold),
  SUM(final_price * units_sold) AS
   total_revenue...
GROUP BY category
ORDER BY total_revenue DESC;
```

Toys	**₹506 Billion**
Beauty	**₹503 Billion**
Fashion	**₹501 Billion**

Business Value: These three pillars alone account for **>₹1.5 Trillion** in GMV, dictating primary inventory resource allocation.

Query-to-Insight Card - Brand Dominance

```
SELECT brand,
  AVG(discount_percent),
  SUM(total_revenue)...
LIMIT 10;
```

Adidas	**₹271 Billion**
Nike	**₹270 Billion**

Business Value: Top brands dominate via a highly disciplined, consistent ~21% average discount strategy, avoiding a race to the bottom.

Step 8 - Advanced SQL Business Queries (Part 2)

Engine Room

Executive Terminal

Query-to-Insight Card - Consumer Price Tiers

```sql
CASE WHEN final_price BETWEEN 20001 AND 50000 THEN 'Premium'
     WHEN final_price > 50000 THEN 'Luxury'
     WHEN final_price <= 20000 THEN 'Budget' END AS price_tier,
SUM(final_price * units_sold) AS tier_revenue,
COUNT(product_id) AS product_count
GROUP BY price_tier
ORDER BY tier_revenue DESC;
```

Premium Tier: **₹2.77 Trillion** Revenue across 33,842 products.

Business Value: The Premium Tier (₹20k-₹50k) acts as the undisputed anchor for Gross Merchandise Value, dramatically outperforming budget and luxury tiers.

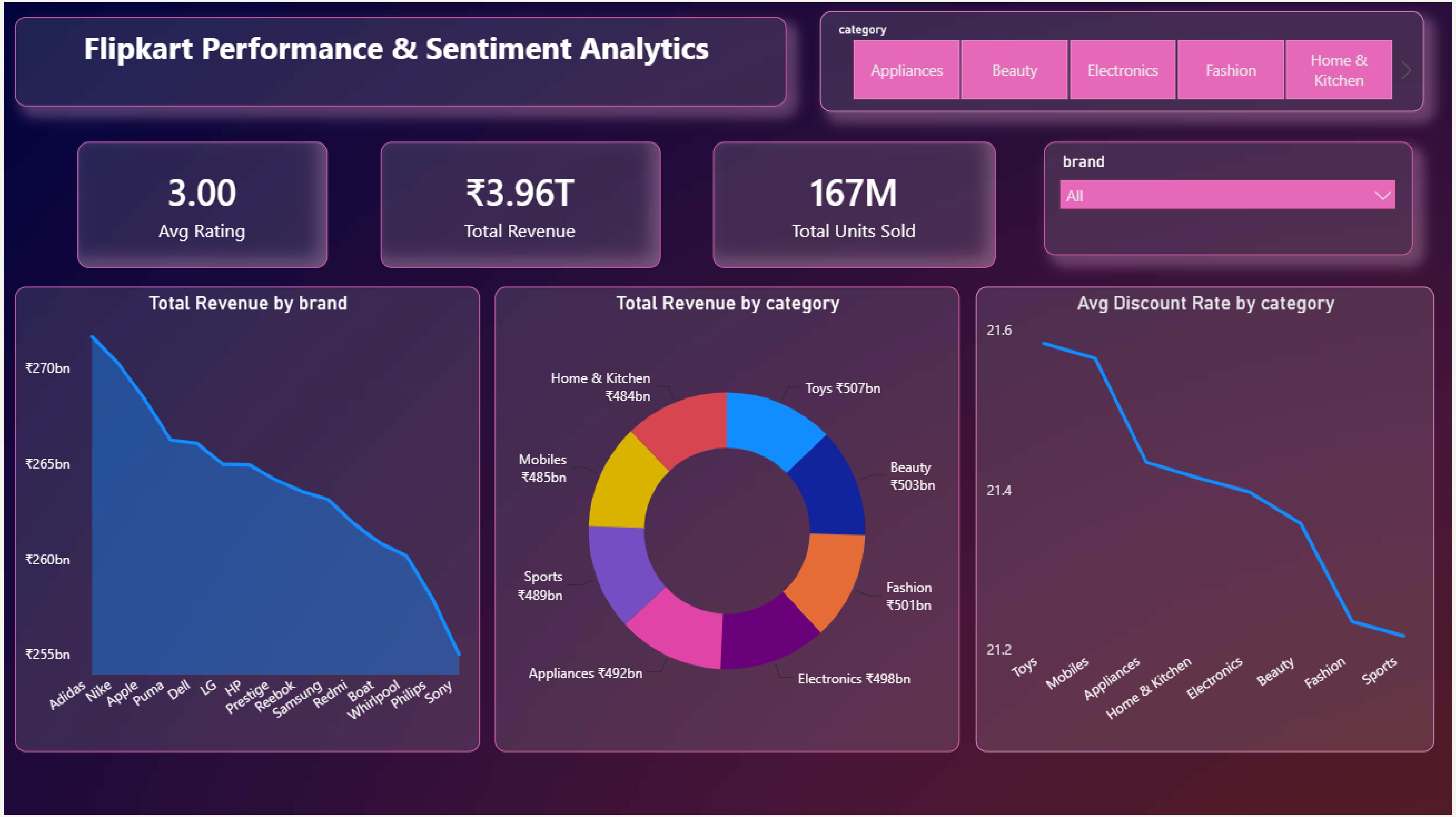
Query-to-Insight Card - Quality Risks

```sql
SELECT product_id, brand, rating, units_sold,
       SUM(final_price * units_sold) AS total_revenue
FROM flipkart_cleaned
WHERE rating < 3.0 AND units_sold > 2000
ORDER BY units_sold DESC;
```

Dell FKP0030096	Rating: 1.2	Units: 4,999	Rev: ₹218M
Whirlpool FKP0040699	Rating: 1.9	Units: 4,999	

Business Value: Identifies toxic SKUs driving massive volume but severe customer churn risk. Immediate QA intervention required.

Step 9 - Interactive Dashboard Integration (Power BI)



Step 10 - Strategic Conclusions & Business Outcomes

Executive Terminal

1. Pricing Strategy

The ~21% discount mark is the optimal equilibrium for top brands (**Adidas, Apple, Nike**). Aggressive discounting beyond 30% yields diminishing revenue returns.

2. Inventory Focus

Resource allocation must aggressively prioritize the the **Premium Tier** (**₹20k-₹50k**), which serves as the core engine for platform GMV.

3. Seller Audits (Action Required)

Immediate QA intervention required for the high-volume, low-rating (<2.5) products in the bottom-right matrix quadrant to protect brand sentiment and reduce return rates.

Data pipeline successfully processed 66,516 rows, yielding actionable insights modeling **₹3.9+ Trillion** in platform GMV.